CS230: Introduction to Machine Learning

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Course Description

This course provides an introduction to machine learning, covering fundamental concepts, techniques, and applications. Students will learn about data preprocessing, supervised and unsupervised learning, model evaluation, and basic neural networks. The course includes hands-on projects using Python and scikit-learn. The course will prepare students for advanced topics in deep learning, which will be covered in a subsequent course.

Prerequisites

- Basic knowledge of Python programming
- Basic understanding of linear algebra, probability, and statistics

Credit Hours: 4

Textbook

- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurlien Gron
- *Probabilistic Machine Learning*, by Kevin Patrick Murphy
  Download Link

Course Objectives

1. To understand the fundamental concepts of machine learning and its applications.
2. To learn how to preprocess data for machine learning models.
3. To explore various supervised and unsupervised learning algorithms.
4. To evaluate and select appropriate models for different types of data and tasks.
5. To implement machine learning algorithms using Python and scikit-learn.
6. To gain an introductory understanding of neural networks and their training.

Course Outcomes

By the end of this course, students will be able to:

1. Explain the basic concepts and types of machine learning.
2. Preprocess and prepare data for machine learning tasks.
3. Implement and apply various supervised learning algorithms.
4. Implement and apply various unsupervised learning algorithms.
5. Evaluate and tune machine learning models using appropriate metrics and techniques.
6. Develop basic neural network models and understand their training processes.
7. Apply machine learning techniques to real-world problems through a comprehensive final project.

Course Structure

Week 1: Introduction to Machine Learning

- What is Machine Learning?
- Types of Machine Learning: Supervised, Unsupervised, Semi-supervised, Reinforcement Learning
- Applications of Machine Learning
- Course Logistics and Python Setup

Week 2: Python for Machine Learning

- Python Libraries: NumPy, pandas, matplotlib
- Introduction to scikit-learn
- Basic Data Structures and Manipulations

Week 3: Data Preprocessing

- Data Cleaning and Preparation
- Handling Missing Values
- Feature Scaling and Normalization
- Data Splitting: Training and Test Sets

Week 4: Supervised Learning: Regression

- Linear Regression
- Polynomial Regression
- Evaluation Metrics: MSE, RMSE, MAE
- Practical Session: Implementing Regression Models

Week 5: Supervised Learning: Classification

- Logistic Regression
- k-Nearest Neighbors (k-NN)
- Evaluation Metrics: Accuracy, Precision, Recall, F1-Score
- Practical Session: Implementing Classification Models

Week 6: Model Evaluation and Selection

- Cross-Validation
- Hyperparameter Tuning: Grid Search and Random Search
- Bias-Variance Tradeoff

Week 7: Supervised Learning: Support Vector Machines

- SVM Concepts
- Kernel Trick
- Practical Session: Implementing SVM Models

Week 8: Supervised Learning: Decision Trees and Ensemble Methods

- Decision Trees
- Random Forests
- Gradient Boosting Machines
- Practical Session: Implementing Ensemble Models

Week 9: Midterm Exam and Project Proposal

- Midterm Exam
- Discussion of Final Project Requirements
- Submission of Project Proposals

Week 10: Unsupervised Learning: Clustering

- k-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Practical Session: Implementing Clustering Algorithms

Week 11: Unsupervised Learning: Dimensionality Reduction

- Principal Component Analysis (PCA)
- t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Practical Session: Implementing Dimensionality Reduction Techniques

Week 12: Unsupervised Learning: Association Rule Learning

- Apriori Algorithm
- Eclat Algorithm
- Practical Session: Implementing Association Rule Learning

Week 13: Anomaly Detection and Recommendation Systems

- Anomaly Detection Techniques
- Collaborative Filtering
- Content-Based Filtering
- Practical Session: Building a Recommendation System

Week 14: Introduction to Neural Networks

- Perceptrons and Multi-Layer Perceptrons (MLP)
- Activation Functions
- Forward and Backward Propagation
- Practical Session: Implementing a Basic Neural Network with Keras

Week 15: Neural Network Training and Optimization

- Loss Functions
- Gradient Descent and Optimization Algorithms
- Overfitting and Regularization: Dropout, L2 Regularization
- Practical Session: Training Neural Networks

Week 16: Final Project Presentations

- Final Project Submission and Presentation
- Course Review and Future Directions in Machine Learning

The weekly coverage might change as it depends on the progress of the class. However, you must keep up with the earlier lectures and reading assignments.

Course Structure

- **Lectures + Lab**: 4 hours per week
- **Quizzes**: Bi-weekly. Five or more.
- **Midterm Exam**
- **Final Exam**
- **Final Project and Presentation**

Assignments

- Weekly coding assignments to implement the concepts covered in lectures.
- Bi-weekly quizzes to assess understanding of theoretical concepts.
- Final project: A comprehensive project where students apply machine learning techniques to a real-world problem and present their findings.

Evaluation

- Quiz and Assignments: 20%
- Final Project: 25%
- Midterm Exam: 20%
- Final Exam: 30%
- Participation and Attendance: 5%

Additional Resources

- Scikit-Learn Documentation
- Keras Documentation
- Coursera Machine Learning Course by Andrew Ng
- Google Machine Learning Crash Course